

THE DANCE OF THE NETWORK – FROM QUANTUM FOAM TO SPIRAL DISC

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ABSTRACT

In VG4, we introduced the flattened geometry of the spiral-matrix and showed that flattening is a fundamental property of the Network, not just an effect of the closing phase. In VG5, we step back – or rather, step forward toward the beginning. We describe in detail how the Network unfolds from the **pre-inflationary state** (quantum foam of relations), going through the explosive release phase (inflation) and expansion, up to the moment when **nodal tension** reverses its sign and triggers contraction. We explain how the **subquantum wall** and the **memory of cycles** make it possible to return to the original pattern without breaking the laws of physics. VG5 is, in essence, a **complete choreography of the cosmic dance** – from foam to disc, from disc to foam.

Keywords: quantum foam of relations, maximal relational density, nodal tension, subquantum wall, nodal fragments, inflation, contraction, spiral disc, VG.

1. QUANTUM FOAM OF RELATIONS – THE PRE-INFLATIONARY STATE

Before any opening, the Network exists in a state we have called the **quantum foam of relations**. It is not a singularity in the classical physics sense, but a state of **maximal relational density**: all nodes are perfectly aligned along the flattened spiral-ribbon, distances between them are zero, and couplings are at maximal intensity, but only between immediate neighbors. In this state, **space and time do not yet exist** – they are only pure potentialities. It is the state of **perfect coherence** from which everything will emerge.

This quantum foam is not chaotic. It has a **pattern** – one that will determine all subsequent evolution: the flattened logarithmic spiral, with all nodes having their spins aligned in the same direction. This pattern is the **memory** of the previous cycle, preserved at the level of the subquantum wall.

2. THE THRESHOLD FLUCTUATION AND THE INFLATIONARY EXPLOSION

At some point, a **quantum threshold fluctuation** – a local „jitter” in the foam of relations – disturbs the perfect alignment. This is the **spark** that triggers the release of accumulated nodal tension: the entire Network **explosively unfolds** in an exponential expansion (inflation). This is the **first moment of time** – because, from this point onward, a clear succession between „before” and „after” appears. And it is the first moment of space – because, with the unfolding, distances between nodes become finite, and the couplings J_{ij} begin to generate a real **metric**.

In this phase, **nodal tension** drops sharply but does not disappear completely. It remains as a **residual tension** – what modern physics calls the cosmological constant. This residual tension will continue to influence all subsequent evolution.

3. EXPANSION AND THE ROLE OF THE SUBQUANTUM WALL

After the inflationary phase, a **slower expansion** follows, dominated by gravity and residual nodal tension. As nodes move apart, couplings weaken, and global coherence decreases.

Enter the **subquantum wall** – an emergent property of the Network, manifesting as a **threshold field** that surrounds the Network like an airtight alveolus, without fissures. This wall is inactive during the expansion phase, but as coherence decreases, it begins to **accumulate** energy and information that would otherwise be lost. It behaves as a **memory buffer** between cycles.

What does the subquantum wall contain?

Inside it, in a state of „waiting,“ are **nodal fragments**. These are **not particles** in the standard physics sense (such as the hypothetical tachyons, with their historical baggage). In RTS, nodal fragments are **coupling structures** – patterns of the Network that are not subject to the usual spatio-temporal constraints. They exist in a state of pure potential, without moving through space, and are activated only at the moment of transition between cycles.

Their role in the cosmic cycle:

When activated – at the closing or opening of a cycle – the subquantum wall uses nodal fragments to **transport instantaneously** energy and information between the layers of the Network, ensuring the **continuity of cosmic memory** beyond the singularity.

4. REVERSAL – WHEN NODAL TENSION CHANGES SIGN

At some point, the energy density of expansion becomes smaller than the energy density of residual nodal tension. This is the **critical point**. At this moment, nodal tension – which until then acted as a repulsive force (accelerating expansion) – reverses its sign and becomes an **attractive force**.

This reversal is not abrupt, but **gradual**. It marks the beginning of the **contraction phase** – nodes begin to approach each other again. The subquantum wall, which had accumulated energy, begins to **release** it back into the Network, contributing to the restoration of coherence. Contraction accelerates as couplings become stronger.

5. RE-WINDING – RETURN TO THE QUANTUM FOAM

As contraction progresses, the Network **re-winds** onto the flattened spiral-ribbon, following the same pattern it had at the beginning, but with an **additional thickness** – a new layer of information added to the spiral-matrix. This new layer is the **memory of the cycle** that has just taken place.

In the end, the Network returns to the **quantum foam of relations** state, with all nodes again aligned, distances zero, and couplings at maximal intensity. The only difference is that the spiral-ribbon now has a **finite thickness**, made up of the superimposed layers of all previous cycles. This is the **cosmic information archive** – the spiral-matrix itself.

6. CONCLUSION

VG5 describes the **complete dance** of the Network: foam → opening → inflation → expansion → contraction → re-winding → foam. In this dance, **nodal tension** is the motor, the **subquantum wall** and **nodal fragments** are the memory, and the **flattened spiral** is the choreography. In VG6, we will use this understanding to show how Bousso's critique is **eluded** in RTS – because, in this model, the „inevitable singularity” is, in fact, a **transition gateway** through which the Network passes, leaving behind only subtle imprints.